

# Hypercomplex Graph Neural Network: Towards Deep Intersection of Multi-modal Brain Networks

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**Abstract**—The multi-modal neuroimage study has provided insights into understanding the heteromodal relationships between brain network organization and behavioral phenotypes. Integrating data from various modalities facilitates the characterization of the interplay among anatomical, functional, and physiological brain alterations or developments. Graph Neural Networks (GNNs) have recently become popular in analyzing and fusing multi-modal, graph-structured brain networks. However, effectively learning complementary representations from other modalities remains a significant challenge due to the sophisticated and heterogeneous inter-modal dependencies. Furthermore, most existing studies often focus on specific modalities (e.g., only fMRI and DTI), which limits their scalability to other types of brain networks. To overcome these limitations, we propose a HyperComplex Graph Neural Network (HC-GNN) that models multi-modal networks as hypercomplex tensor graphs. In our approach, HC-GNN is conceptualized as a dynamic spatial graph, where the attentively learned inter-modal associations are represented as the adjacency matrix. HC-GNN leverages hypercomplex operations for inter-modal intersections through cross-embedding and cross-aggregation, enriching the deep coupling of multi-modal representations. We conduct a sta-

tistical analysis on the saliency maps to associate disease biomarkers. Extensive experiments on three datasets demonstrate the superior classification performance of our method and its strong scalability to various types of modalities. Our work presents a powerful paradigm for the study of multi-modal brain networks.

**Index Terms**—Multi-modal graph neural network, neuroimage, hypercomplex-GNN, tensor, brain

## I. INTRODUCTION

The pursuit of decoding brain networks has provided us with insights into the neural mechanisms of the brain and the intricate relationships between individual brain dysfunctions and behavioral phenotypes [1], [2]. Recent investigations have made significant progress in examining and identifying connectome patterns associated with brain disorders such as autism spectrum disorder (ASD) and attention-deficit/hyperactivity disorder (ADHD) [3]–[5]. In this line of works, Graph Neural Networks (GNNs) have been extensively utilized in connectome-wide association studies and have emerged as state-of-the-art tools for exploring the topological and spatial organization of brain networks.

The integration of multi-modal networks has emerged as a significant area of research in the study of brain networks. These networks encompass functional networks derived from functional magnetic resonance imaging (fMRI) and electroencephalography (EEG), as well as structural networks constructed from diffusion tensor imaging (DTI) or diffusion spectrum imaging (DSI) [6], [7]. Exploring the relationships between different modalities enables the capture of important complementary information and the enhancement of representations. Previous studies suggest that multiple types of brain connectivity might be mediated by each other [8]. Furthermore, multi-modal studies allow for the exploration of brain states by examining neuron activation and connections in vivo, thus providing a more comprehensive understanding with distinct biomarkers [9], [10]. Accordingly, combining multi-modal information for brain network studies is both natural and well-justified.

One of the challenges in multi-modal brain network studies is that the multiple modalities often have complex and heterogeneous structures. Simply applying feature embedding to the entire feature set is inefficient for capturing meaningful representations across all modalities. Additionally, it remains nontrivial to effectively learn from each modality to enrich representations. Moreover, most existing models [5], [11], [12]

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are designed for specific brain networks and lack scalability to other types. For example, there are various covariance matrix parameterizations for constructing brain networks, such as Pearson correlation, partial correlation, and the tangent space of covariance matrices. These contribute to diverse tendencies in different disease analyses [13], [14]. Consequently, integrating these distinctive types of brain networks provides a more comprehensive manner to uncovering distinctive biomarkers. However, most existing approaches fail to flexibly couple various types or modalities.

A straightforward and natural approach to exploring multi-modal data is to model it as multi-dimensional arrays, known as tensors. Tensors have been proven to be powerful and promising tools for uncovering complex underlying structures in data, enabling the separation of common and independent components [15]. Most tensor-based methods employ tensor decomposition techniques, such as Canonical Polyadic (CP) decomposition and Tucker Decomposition (TD), to identify key features and enhance learning efficiency [16]. However, these approaches often have limitations in terms of lossy transformations and may overlook important intrinsic complementary information, potentially leading to suboptimal results.

In this regard, we propose a HyperComplex Graph Neural Network (HC-GNN) for multi-modal brain network learning. This approach enriches features and facilitates deep interactions of representations through cross-embedding and cross-aggregation in an end-to-end manner. Specifically, we construct tensor graphs by formulating multimodal features as tensors and further implement a hypercomplex neural network (HNN) [17], [18] to encode tensor representations. The approach involves constructing composite tensors to learn latent multimodal representations and generating high-order generalizations of matrices to capture the consistency among different modalities. Furthermore, the deep intersection inherent in HNN allows for richer representations by expressing neuron outputs with multiple indices.

In addition, we introduce a dynamic mechanism to capture the inter-modal dependencies by relating attentive multi-modal dependencies as the adjacency matrix. This approach adaptively characterizes the intrinsic associations between different modalities and addresses the issue of unknown node relationships in multi-modal graphs. Furthermore, most existing multi-modal networks overlook the importance of neuronal synchrony, which involves information gating and dynamic binding of inter-modal representations. To this end, we introduce a gating mechanism to control the information flow and dynamically bind inter-modal aggregations attentively. Extensive experiments on three real-world datasets demonstrate that our proposed method is a superior tool for multi-modal brain network representation learning. It outperforms other baselines in terms of prediction performance, generalizability, and scalability.

To summarize, our contributions are as follows:

- We propose the HC-GNN network for multi-modal brain network analysis, which enables the modeling of various modalities in a tensor graph. A gated graph convolution is introduced to attentively regularize inter-modal intersections.

- HC-GNN leverages hypercomplex operations for inter-modal intersections through cross-embedding and cross-aggregation, thereby enriching the deep coupling of multi-modal representations.
- Results on model interpretability demonstrate that HC-GNN is capable of locating meaningful disease patterns across various modalities. Our findings regarding the precuneus provide additional evidence that ASD and ADHD might fall on the same continuum and share similar patterns.

This paper extends the preliminary version of Tensor-based Complex-values GNN (TC-GNN) [19] by extending the model from two modalities to multiple modalities with hypercomplex representations to facilitate dynamic interactions. We also improved the overall framework by introducing residual connections and edge-to-node convolution to enhance efficiency and effectiveness. Extended experiments and biological interpretations are also provided to better understand how our model facilitates multi-modal brain network analysis.

## II. RELATED WORKS

### A. Brain Connectome Study

Recently, deep learning approaches have prompted a shift in connectome studies towards subject-level predictions with promising accuracy. These approaches include Convolutional Neural Networks (CNNs) [20]–[22], and Graph Neural Networks (GNNs) [12], [23]. GNN approaches are more effective than CNN methods by capturing the structural information and topological properties of the brain graph. For example, BrainGNN [12] proposes ROI-aware graph convolutional layers and ROI-selection pooling layers for brain network analysis. [24] proposed a multi-graph attention network (MGAT) with bilinear convolution for diagnosis. GC-GAN is proposed for optimizing the graph topology and capturing intricate connectome patterns [25]. Although the connectome, incorporating the strength of connections in their edges, assumes a graph-like organization, the connections are limited in depicting sophisticated non-linear relationships between brain regions [26]. Applying the derived functional connectivity as the adjacency matrix, which is measured linearly between two brain regions, fails to model complex associations between brain regions. To address the issue of unclear graph structure, DGCNN dynamically gathers nearby information to recover topology for learning [27]. [26] suggests learning the graph structure based on the small-world model. [28] aggregates the top- $k$  nearest nodes to update node features. Based on these studies, we employ a dynamic mechanism to model attentive dependencies between brain regions for multi-modal graphs.

### B. Multi-modal brain network learning

Multi-modal neuroimaging technologies have provided unprecedented opportunities for disease diagnosis and connectome-behavior association studies. Deep learning methods are feasible for capturing more meaningful features, contributing to a better understanding of the brain network. For example, Wang et al. [21] proposed the use of a multi-channel convolution kernel for the fusion of functional and structural

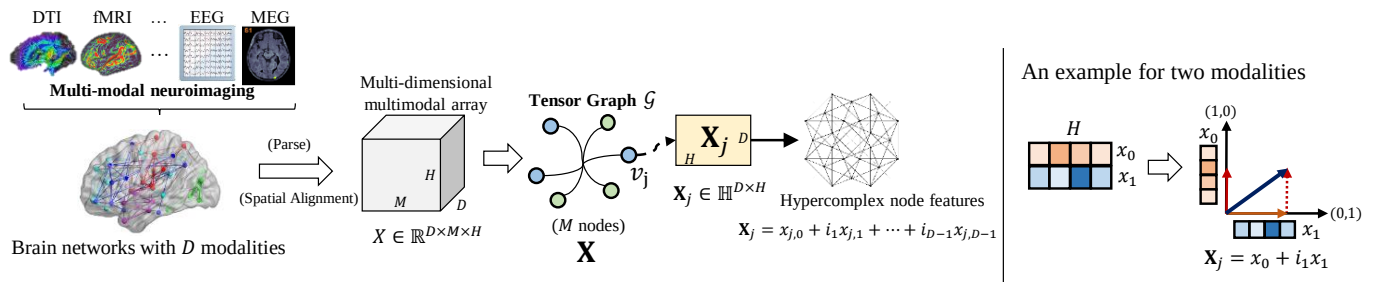


Fig. 1. Tensor graph construction. First, we perform spatial alignment to map multi-modal brain networks into the same space. Next, we formulate the concatenated multi-modal multi-dimensional array as a tensor graph. In this graph, there are  $M$  nodes, with each node represented as a tensor.

brain networks. Furthermore, well-established graph-based models can effectively capture multi-modal representations. For instance, Huang et al. [29] constructed a graph with variational edges incorporating multimodal features. M-GCN introduces to integrates complementary information from rs-fMRI and DTI by graph Laplacian [11]. Ji et al. [30] applied graph hashing learning to preserve the spatial relationships. In addition, the use of hypergraphs provides a strategy for capturing dependencies among nodes by constructing hyperedges based on each modality. Examples include Hypergraph Neural Networks (HNN) [31] and Dynamic Hypergraph Neural Networks (DHGNN) [32]. Additionally, [33] proposed a graph parser and matcher to capture the correspondence between nodes from different modalities. In another approach, MMP-GCN proposes a dual-modality fused brain connectivity network in a population graph [34]. The Triplet Attention Network is proposed for the high-order feature extraction with the attention mechanism [35]. However, many of these studies either overlook important complementary information between modalities or lack scalability to handle data with more modalities.

### C. Hypercomplex and Complex neural networks

Recently, there has been increasing attention on Hypercomplex neural networks (HNNs) in signal processing. HNNs offer a new perspective for enhancing representational capacity by utilizing hypercomplex numbers instead of the commonly used real-valued numbers. By leveraging the interactions among the real and imaginary units, HNNs can capture internal relationships in inputs and preserve preexisting correlations [18], [36]. In recent years, there has been a growing body of research exploring the potential use of hypercomplex neural networks in various domains. For instance, [37] applied complex neural networks for object detection in images, with depth sensors represented as the imaginary part. [18] introduced a family of parameterized HNNs and their applications in multi-dimensional images. [38] developed a quaternion convolutional neural network for automatic speech recognition, and [39] utilized hypercomplex context for the scene graph generation. However, there have been few hypercomplex neural networks studies investigating the potential use of multi-modal learning. In this work, we address multi-modal representation learning by formulating multi-modal data as hypercomplex tensors, which is extended from our previous study [19].

## III. METHOD

### A. Problem Definition

We illustrate the process of our method in Fig. 1. To begin, we convert the pre-processed brain network arrays into tensors for each participant. We further construct a hypercomplex tensor graph  $\mathcal{G}$  by mapping the multi-dimensional node features to hypercomplex numbers. Our overarching objective in this study is to discern the presence of disease or phenotype through a multi-modal graph classification task, accomplished through a mapping function denoted as  $f_{\mathcal{G}} : \mathcal{G} \rightarrow \mathcal{Y}$ . In this study, hypercomplex numbers are highlighted in bold, and the arrays are denoted with capital alphabets.

**Brain Network Graph:** Brain networks are constructed from neuroimages or neuron signals by aligning the input with a template consisting of  $M$  regions. These brain networks are denoted as  $X_s \in \mathbb{R}^{M \times M}$ . Each element,  $x_{i,j}$  of matrix  $X_s$ , represents the covariance or connectivity strength between the  $i$ -th and  $j$ -th regions of interest (ROI). For multi-modal brain networks, we initially map pre-processed brain networks onto the same space with an identical number of ROIs, which is  $M$ . When presented with an input  $X = \{X_0, X_1, \dots, X_{D-1}\}$ ,  $X \in \mathbb{R}^{D \times M \times M}$  comprising  $D$  modalities, our first step is to parse each modality into  $H$  features by an encoder, where  $X_d \in \mathbb{R}^{M \times H}$  for each modality  $d$  in the set  $\{0, 1, \dots, D-1\}$ .

**Hypercomplex Tensor Graph:** We took the integration of multi-modal features a step further by constructing a hypercomplex tensor graph denoted as  $\mathcal{G}$ . This graph comprises  $M$  nodes, and each node is portrayed as an array  $X_u \in \mathbb{R}^{H \times D}$ , with  $u$  belonging to the node set  $\mathcal{V}$ . The features of each modality are expressed either in the real part or as a hyperimaginary unit. To illustrate this concept, consider the example in Fig. 1, which involves two modalities. In this scenario, one modality is depicted as the real part, while the other is represented as the imaginary part.

### B. Hypercomplex neural networks

In this section, we would like to introduce the hypercomplex operations in neural networks for multimodal studies.

**1) Hypercomplex addition and multiplication:** For a basis of  $\{1, i_1, \dots, i_{D-1}\}$ , the multi-modal input  $X = \{X_0, X_1, \dots, X_{D-1}\}$  with  $D$  modalities is formulated as a hypercomplex number  $\mathbf{X}$ :

$$\mathbf{X} = X_0 + i_1 X_1 + \dots + i_{D-1} X_{D-1}, X_d \in \mathbb{R}^{M \times H} \quad (1)$$



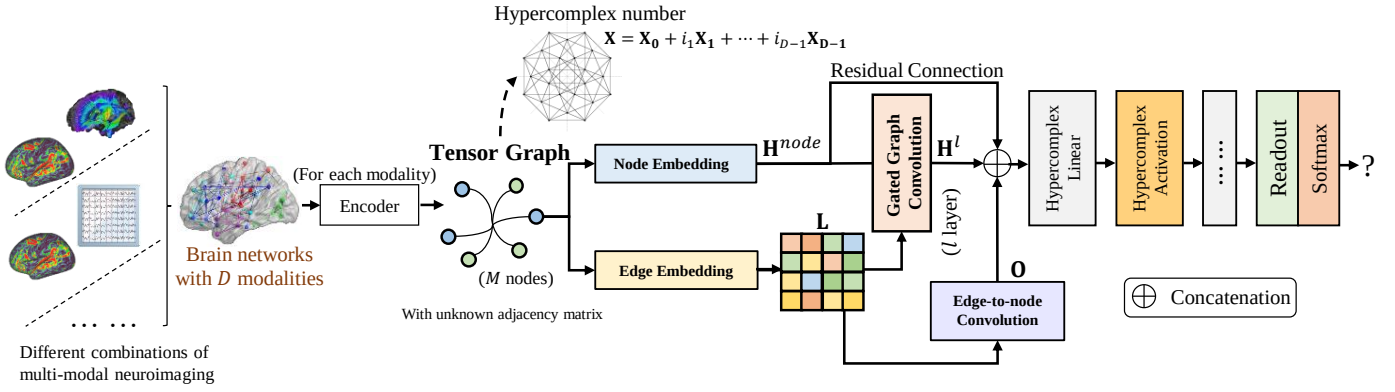


Fig. 2. Illustration of the proposed HC-GNN, with hypercomplex node embedding, hypercomplex edge embedding, and edge-to-node convolution. HC-GNN is feasible to model different combinations of multi-modal neuroimaging. An encoder is employed to extract deep representations for each modality. These modality-specific features are then utilized to construct the hypercomplex nodes.

The addition of two hypercomplex numbers  $\mathbf{X} = X_0 + i_1 X_1 + \dots + i_{D-1} X_{D-1}$  and  $\mathbf{Y} = Y_0 + i_1 Y_1 + \dots + i_{D-1} Y_{D-1}$ ,  $Y_d \in \mathbb{R}^{M \times H}$  is obtained by a component-wise manner as:

$$\mathbf{X} + \mathbf{Y} = (X_0 + Y_0) + i_1 (X_1 + Y_1) + \dots + i_{D-1} (X_{D-1} + Y_{D-1}) \quad (2)$$

The product between  $\mathbf{X}$  and  $\mathbf{W} = W_0 + i_1 W_1 + \dots + i_{D-1} W_{D-1}$ ,  $W_d \in \mathbb{R}^{H \times K}$ , denoted by the juxtaposition of  $\mathbf{X}$  and  $\mathbf{Y}$ , is given by:

$$\begin{aligned} \mathbf{XW} &= (X_0 W_0 + \sum_{u,v=1}^{D-1} X_u W_v a_{uv,0}) \\ &+ i_1 (X_0 W_1 + X_1 W_0 + \sum_{u,v=1}^{D-1} X_u W_v a_{uv,1}) + \dots \\ &+ i_{D-1} (X_0 W_{D-1} + X_{D-1} W_0 + \sum_{u,v=1}^{D-1} X_u W_v a_{uv,D-1}) \end{aligned} \quad (3)$$

Here,  $a_{uv}$  is obtained from a multiplication table determined by the hypercomplex numbers in the intersection:

$$i_u i_v = a_{uv,0} + i_1 a_{uv,1} + \dots + i_{D-1} a_{uv,D-1}, u, v \in \{1, \dots, D-1\} \quad (4)$$

For example, we provide a multiplication table for  $D = 4$  in Table I. As an illustration, the product of  $i_1$  and  $i_2$  corresponds to element  $i_3$  in the 2nd column and 3rd row.

TABLE I  
EXAMPLE OF HYPERCOMPLEX MULTIPLICATION TABLE FOR  $D = 4$ .

|       | 1     | $i_1$ | $i_2$  | $i_3$  |
|-------|-------|-------|--------|--------|
| 1     | 1     | $i_1$ | $i_2$  | $i_3$  |
| $i_1$ | $i_1$ | -1    | $i_3$  | $-i_2$ |
| $i_2$ | $i_2$ | $i_3$ | -1     | $i_1$  |
| $i_3$ | $i_3$ | $i_2$ | $-i_1$ | -1     |

**2) Hypercomplex convolution:** When working with a hypercomplex filter matrix  $\mathbf{W} = \mathbf{W}_0 + i_1 \mathbf{W}_1 + \dots + i_{D-1} \mathbf{W}_{D-1}$ , the convolution applied to an input  $\mathbf{X} = \mathbf{X}_0 + i_1 \mathbf{X}_1 + \dots + i_{D-1} \mathbf{X}_{D-1}$  is represented as the product of  $\mathbf{W}$  and  $\mathbf{X}$ .

**3) Hypercomplex activation function:** Activation functions play a crucial role in identifying meaningful nonlinear representations. To satisfy the Cauchy-Riemann equation,  $\mathbb{H}\text{ReLU}$  is derived from  $\mathbb{C}\text{ReLU}$  for hypercomplex numbers as follows:

$$\mathbb{H}\text{ReLU}(\mathbf{X}) = \text{ReLU}(X_0) + i_1 \text{ReLU}(X_1) + \dots + i_{D-1} \text{ReLU}(X_{D-1}) \quad (5)$$

$\mathbb{H}\text{ReLU}$  facilitates nonlinear activation independently on both the real and multiple imaginary parts.

**4) Hypercomplex Readout:** A readout function is to convert the hypercomplex numbers into real by  $\text{Readout}(\mathbf{X}) = \sqrt{X_0^2 + X_1^2 + \dots + X_{D-1}^2}$ .

Hypercomplex neural networks provide the capacity to model multi-modal inputs by harnessing various imaginary units. For example, the quaternion domain (with  $D = 4$ ) is well-suited for handling 3-modality or 4-modality data, whereas the octonion domain is more appropriate for accommodating a higher number of modalities. In practical applications, complex and quaternion neural networks can effectively address most multi-modal learning cases.

### C. HC-GNN

Leveraging hypercomplex operations, we introduce hypercomplex graph convolution to gather inter-region information and enable graph convolution on tensor graphs. The structure of our proposed Hypercomplex Graph Neural Network (HC-GNN) is visually depicted in Fig. 2. Here's how it works: Multimodal inputs are encoded and then combined into a hypercomplex tensor denoted as  $\mathbf{X}$ . Additionally, we employ a dynamic mechanism for edge embedding, which leads to the construction of a dynamic graph. Following this, we apply gated hypercomplex graph convolution to capture multi-modal topological information by considering feature interactions within the graph domain. Finally, the resulting aggregated features are passed through a hypercomplex Multi-Layer Perceptron (MLP) and read out as real numbers for the classification task.

1) *Hypercomplex node embedding*: The multi-modal node features are aggregated from the connection features to the centroid node with a hypercomplex multi-layer perception embedding. The output features  $\mathbf{H}^{node} \in \mathbb{H}^{M \times H}$  are obtained by a weighted sum. In detail, given a vertex  $v_j$ , the corresponding features  $\mathbf{h}_j^{node} \in \mathbb{H}^H$  are obtained from its connections:

$$\begin{aligned} \mathbf{h}_j^{node} &= f_n(\mathcal{N}_j) = \sum_{k=0}^{H-1} \mathbf{w}_k * \mathbf{x}_{jk} \\ &= \sum_{k \in \mathcal{N}(j)} (w_{k,0} + i_1 w_{k,1} + i_2 w_{k,2} + \dots + i_{D-1} w_{k,D-1}) * \\ &\quad (x_{jk,0} + i_1 x_{jk,1} + i_2 x_{jk,2} + \dots + i_{D-1} x_{jk,D-1}) \end{aligned} \quad (6)$$

where  $\mathbf{w}_k$  is a hypercomplex learnable parameter.

2) *Dynamic hypercomplex edge embedding*: Typically, edge connections are conditioned on the node field. However, in the context of brain network studies, the derived brain connectivity predominantly reflects linear associations. As a first step, we apply a linear transformation to map the hypercomplex input into hidden features using learnable parameters  $\mathbf{w}^e$ . This transformation is expressed as:

$$\hat{\mathbf{L}}_{h,j,k} = \sum_{m=1}^M (\mathbf{w}_{m,h}^{e1} \mathbf{x}_{j,m} + \mathbf{w}_{m,h}^{e2} \mathbf{x}_{m,k}), \hat{\mathbf{L}} \in \mathbb{H}^{H^{edge} \times M \times M} \quad (7)$$

Here,  $H^{edge}$  represents the hidden size of the edge embedding features. Furthermore, the dynamic adjacency matrix is generated through a linear mapping followed by a sigmoid function  $\sigma$ . Specifically, each element  $\mathbf{L}_{j,k}$  in the  $j$ -th column and the  $k$ -th row of matrix  $\mathbf{L}$  is defined as:

$$\mathbf{L}_{j,k} = \sigma \sum_{h=1}^{H^{edge}} \sum_{m=1}^M (\mathbf{w}_{m,h}^{e3} \hat{\mathbf{L}}_{h,j,m} + \mathbf{w}_{m,h}^{e3} \hat{\mathbf{L}}_{h,k,m}), \mathbf{L} \in \mathbb{H}^{M \times M} \quad (8)$$

In these equations, we manipulate the hypercomplex domain for edge and adjacency representations to capture the brain connectivity characteristics.

3) *Edge-to-node convolution*: The Edge-to-Node (E2N) convolution, as proposed in BrainNetCNN [20], takes an adjacency matrix from each feature map as input and generates a vector as output. In the context of hypercomplex operators, the output, denoted as  $\mathbf{O}$ , of an E2N layer is computed as follows:

$$\mathbf{O}_{h,j} = \sum_{h=1}^{H^{edge}} \sum_{m=1}^M (\mathbf{w}_{m,h}^{o1} \hat{\mathbf{L}}_{j,m} + \mathbf{w}_{m,h}^{o2} \hat{\mathbf{L}}_{m,j}), \mathbf{O} \in \mathbb{H}^{H^{edge} \times M} \quad (9)$$

In this formulation, we employ hypercomplex representations to perform the E2N convolution, aiming to capture intricate features in the brain network data.

4) *Gated Graph Convolution*: In our study, we employ graph convolution to gather inter-regional information and establish topological features for the purpose of classification. To streamline the computations, we specifically employ spatial aggregation. Now, given a learned hypercomplex adjacency matrix  $\mathbf{L} = L_0 + i_1 L_1 + \dots + i_{D-1} L_{D-1}$  and an input

feature  $\mathbf{H} = H_0 + i_1 H_1 + \dots + i_{D-1} H_{D-1}$ , the complex graph convolution on the  $l$ -th layer is computed through a hypercomplex matrix multiplication:

$$\mathbf{H}^l = \text{GCN}(\mathbf{L}, \mathbf{H}^{l-1}; \mathbf{W}^l) = \mathbf{L} \mathbf{H}^{l-1} \mathbf{W}^l \quad (10)$$

It's important to note that  $\mathbf{H}^0 = \mathbf{H}_{node}$ , and  $\mathbf{W}^l$  represents the learnable parameters of the  $l$ -th layer. This formulation allows us to capture the intricate relationships within the hypercomplex graph for our classification task.

Furthermore, while hypercomplex operators excel in computing and fusing multi-modal representations, they tend to overlook the inherent heterogeneity among different modalities. In light of this, we propose to harness gated transformations between various modalities. The gating mechanism, a commonly used tool in neural networks, is designed to enhance the learning of long-term relationships and protect the cell state from undesirable updates [40]. In simple terms, the gates regulate the message passing of signals. In our study, we implement this gating mechanism using an attention mechanism, applied as an element-wise product during graph message passing, ultimately influencing the scaling of inter-unit aggregation. This attention mechanism is represented as:

$$\begin{aligned} g_{u,v}^l &= \sigma(W_{uv}^{g1} \cdot Z_u^l + W_{uv}^{g2} \cdot Z_v^l + B_{uv}) \\ \forall u, v &\in \{1, 2, \dots, D-1\} \end{aligned} \quad (11)$$

Here,  $W^{g1}$ ,  $W^{g2}$ , and  $B$  represent the learnable parameters. The values of  $g_{u,v}$  indicate the gating for aggregation between the  $u$  and  $v$  hyperimaginary axes, with  $\sigma$  denoting the sigmoid activation function. Notably, the state transition representations  $\mathbf{Z}^l$  play a significant role, with each element  $\mathbf{Z}_j^l$  determining the  $j$ -th node according to the following equation:

$$\mathbf{Z}_j^l = \|\{\text{Avg}(\mathbf{H}_k^l), \text{Max}(\mathbf{H}_k^l)\}\} \quad (12)$$

Here,  $\|$  represents the concatenation operation, combining the average and maximum values from neighboring nodes to form the transition state representation. Consequently, the gates for inter-axis filtering can be expressed as a gate matrix  $G_\star$  as follows:

$$\begin{bmatrix} 1 & g_{0,1} & \dots & g_{0,D-1} \\ g_{1,0} & 1 & \dots & g_{1,D-1} \\ g_{2,0} & g_{2,1} & \dots & \dots \\ g_{D-1,0} & g_{D-1,1} & \dots & 1 \end{bmatrix}$$

These gate matrices help regulate the flow of information and enhance the network's ability to capture the complex interactions within the hypercomplex graph. We reformulate the gating mechanism into the graph convolution as follows:

$$\mathbf{H}^l = g_\star^l \circ \text{GCN}(\mathbf{L}, \mathbf{H}^{l-1}; \mathbf{W}^l) = (g_\star \circ \mathbf{L} \mathbf{H}^{l-1}) \mathbf{W}^l \quad (13)$$

When dealing with only two modalities (i.e.,  $D = 2$ ), the gated convolution takes the form:

$$\begin{aligned} g_\star^l &\circ \mathbf{L} \mathbf{H}^{l-1} \\ &= L_0 H_0^{l-1} - L_1 H_1^{l-1} + i g_{0,1}^l L_0 H_1^{l-1} + i g_{1,0}^l L_1 H_0^{l-1} \end{aligned} \quad (14)$$

Ultimately, we obtain the encoded features  $\mathbf{H}^l$ ,  $\mathbf{H}_{node}$ , and  $\mathbf{O}$ . These features are concatenated and then passed through

a hypercomplex MLP. The resulting outputs are transformed into real numbers using a readout function, which is employed for classification purposes.

#### D. Biomarker Interpretation

To evaluate the interpretability of our proposed HC-GNN, we utilize saliency maps, a widely adopted tool for biological explanation, to pinpoint critical brain regions. Specifically, we extract activation maps from the aggregated graph feature maps  $\mathbf{S}$  for individual-level analysis. For group-level comparison, we implement the statistical CWAS approach, namely the Multi-Distance Multi-Variance Regression (MDMR) [41], [42]. This comprehensive methodology facilitates a thorough understanding of significant brain regions at group levels. To simulate the null distribution, a random permutation was executed 2,000 times. During each iteration, the pseudo F-statistic score was recalculated. The p-value was then determined by counting how many times the pseudo F-statistics from the permuted values exceeded those obtained from the original data. This process was iteratively applied to all Regions of Interest (ROIs). A p-value of less than 0.05, adjusted after the false discovery rate correction, was considered statistically significant in our experiments.

### IV. EXPERIMENTS

#### A. Datasets

Our proposed HC-GNN is feasible to model various types of brain networks. In this study, three real-world datasets are employed for the evaluation of multiple fMRI functional networks (ABIDE), functional-structural networks (ADNI), fMRI-EEG functional networks (HBN) respectively. The detailed demographic information is listed in Table II with gender (Male/Female), age (Mean $\pm$ Std), and brain network constructions on various datasets.

1) *ADNI Dataset [43]*: We have collected a total of 442 participants including 142 normal controls (NC). And 151 patients were diagnosed as mild cognitive impairment (MCI), and 149 patients diagnosed with Alzheimer's disease (AD). We collect participants with both fMRI and DTI data for evaluating the coupling between functional and structural brain networks during the progression of Alzheimer's disease. To ensure the data integrity and avoid redundancy, we excluded duplicated scans. The subjects were categorized into three groups AD, MCI, and NC based on standard clinical criteria, including assessments such as Mini-Mental State Examination (MMSE) scores and Clinical Dementia Rating (CDR).

2) *ABIDE dataset [44]*: The ABIDE database includes T1 structural brain images, resting-state functional MR images, and phenotypic information from 17 different imaging sites. The initial ABIDE-I dataset consists of 528 individuals diagnosed with autism spectrum disorder (ASD) and 586 controls. We employ the ABIDE dataset to evaluate the performance of HC-GNN in studying multiple types of functional connectomes. These connectomes are constructed by Pearson coefficient correlation, partial correlation, and tangent space of covariance matrices. To further assess the generalization ability of HC-GNN, we implement the ABIDE-II dataset for

testing, which comprises 677 healthy controls and 559 patients with ASD.

3) *HBN dataset [45]*: The HBN dataset was obtained from a cohort of children and adolescents with age ranging from 5 to 21 years. It includes multiple tasks of functional MRI (fMRI) and electroencephalography (EEG) recordings. We specifically extracted the resting-state fMRI and EEG data to construct the corresponding functional connectomes. Our analysis focused on 105 healthy control individuals and 150 patients diagnosed with attention-deficit/hyperactivity disorder (ADHD). We utilize this dataset to investigate the coupling between fMRI and EEG functional connectomes and their relationship with diseases.

#### B. Pre-processing

**fMRI.** All fMRI images underwent pre-processing using the Configurable Pipeline for the Analysis of Connectomes (C-PAC) pipeline [46]. This included several steps such as skull stripping, slice timing correction, motion correction, global mean intensity normalization, nuisance signal regression with 24 motion parameters, and band-pass filtering (0.01-0.08 Hz). The functional images were then registered to the standard anatomical space (MNI152). For region-based analysis, the mean time series for a predefined set of regions of interest (ROIs) were computed and normalized to have zero mean and unit variance. In the ADNI and ABIDE datasets, we used the Schaefer atlas [47] with 100 ROIs to map the fMRI images into parcellations. In the HBN dataset, we utilized the Desikan-Killiany template with 68 ROIs [48] for the cortical cortex learning of both EEG and fMRI data. For the ADNI and HBN datasets, the Pearson correlation coefficient was applied to the parcellated, preprocessed fMRI images to derive functional brain networks. For the ABIDE datasets, the Pearson correlation coefficient, partial correlation, and tangent space of covariance matrices were used to construct three types of functional brain networks.

**DTI.** The DTI images were pre-processed by image denoising, head motion, eddy-current, susceptibility distortion, and field inhomogeneity correction by MRtrix-3 [49]. To reconstruct the fiber tracts, we utilized the 2nd order Integration over Fiber Orientation Distributions method [50], resulting in 10 million streamlines. Subsequently, a Spherical-deconvolution Informed Filtering of Tractograms technique [51] was applied to reduce the streamline count to 5 million, while preserving the essential structural information. The structural connectivity network was constructed based on the number of streamlines connecting each pair of brain regions. In the ADNI dataset, the DTI images were mapped to parcellations using the Schaefer atlas with 100 ROIs [47].

**EEG.** The EEG data were recorded in a sound-shielded room at a sampling rate of 500 Hz, using a bandpass filter of 0.1 to 100 Hz. For each participant, the head circumference was measured, and an appropriately sized EEG net was selected. To improve data quality, the EEG signal underwent pre-processing steps. It was filtered using a FIR filter ranging (8-12) Hz which reflects cognitive and memory performance in particular [52]. Noisy channels were identified and removed,

TABLE II  
CHARACTERISTICS OF PARTICIPANTS OF DIFFERENT DATASETS.

| Dataset  | Number of brain networks | Construction of brain networks                                                                       | Modalities | Group | Gender (M/F) | Age (mean±std) |
|----------|--------------------------|------------------------------------------------------------------------------------------------------|------------|-------|--------------|----------------|
| ADNI     | 2                        | fMRI: Pearson coefficient correlation;<br>DTI: number of fibre connections                           | fMRI, DTI  | NC    | 81/61        | 75.9±9.7       |
|          |                          |                                                                                                      |            | MCI   | 87/64        | 77.0±11.7      |
|          |                          |                                                                                                      |            | AD    | 87/62        | 76.9±8.2       |
| ABIDE-I  | 3                        | fMRI: Pearson coefficient correlation, partial correlation, and tangent space of covariance matrices | fMRI       | NC    | 491/95       | 16.8±7.7       |
|          |                          |                                                                                                      |            | ASD   | 466/62       | 17.0±8.4       |
| ABIDE-II | 3                        | fMRI: Pearson coefficient correlation, partial correlation, and tangent space of covariance matrices | fMRI       | NC    | 429/248      | 14.9±9.6       |
|          |                          |                                                                                                      |            | ASD   | 466/93       | 14.3±8.5       |
| HBN-ADHD | 2                        | fMRI: Pearson coefficient correlation,<br>EEG: Pearson coefficient correlation                       | fMRI, EEG  | NC    | 72/33        | 10.3±3.6       |
|          |                          |                                                                                                      |            | ADHD  | 107/43       | 9.9±3.0        |

and interpolation techniques were applied based on the activity of surrounding channels. To align the EEG data with fMRI space, we performed source reconstruction pre-processing using the Brainstorm software [53]. To analyze the EEG data, the cortical surface was divided into 68 anatomical regions of interest based on the Desikan-Killiany atlas [48]. We focused on the middle 2 minutes of the EEG recording and calculated Pearson correlations by computing correlation coefficients on successive non-overlapping 10-second segments of the EEG time series. The correlation matrices obtained from each segment were averaged to calculate functional connectivity for each participant.

### C. Implementation details

In our implementation, we utilize the GRU as the brain connectome encoder that parses the brain network as  $M$  sequences with  $M$  features into  $H$  features. In different tasks, we obtain different values for  $M$  based on various brain templates, where  $M = 100$  for the ADNI, ABIDE-I, and ABIDE-II datasets, and  $M = 68$  for the HBN datasets. The number of GRU layers is determined through a grid search ranging from 1 to 3.  $H$  is decided in a search of  $\{16, 32, 64, 128\}$ . The parsed brain networks are further constituted into the tensor graph  $\mathbf{X} = X_{fMRI} + iX_{DTI}$  for the ADNI dataset,  $\mathbf{X} = X_{pearson} + i_1X_{partial} + i_2X_{tangent}$  for ABIDE datasets, and  $\mathbf{X} = X_{fMRI} + iX_{eeg}$  for the HBN dataset. The outputs are concatenated and then fed into a 3-layer multi-layer perceptron classifier. This classifier is followed by a  $\mathbb{H}$ ReLU activation function and a dropout layer to enhance model generalization. To train the models, we set the learning rate as  $3e-4$  and the weight decay as  $5e-5$ . All models in this study are trained for 400 epochs, and early stopping is employed if the loss does not decrease for 50 epochs. The training process is performed using PyTorch on a single NVIDIA 2080-Ti GPU. For evaluation, we employ a 10-fold cross-validation strategy, where 10% of the samples are randomly selected for testing in each fold. We assess the performance based on diagnosis accuracy (Acc), sensitivity (Sen), and specificity (Spe).

### D. Competitive methods

In this study, we compare our proposed HC-GNN with the baseline machine learning approaches, the support vector

machine (SVM) and multi-layer perceptron (MLP). We input the upper matrix of the brain networks into these classifiers. For the MLP, we explore different layer settings ranging from 1 to 4. Tensor-based approaches are also compared. We implement the CP and Tucker decomposition approaches for extracting key features, referred to as **TD-MLP** and **CP-MLP**. The number of key feature selections is decided based on a grid search within the range of  $\{16, 32, 64\}$ .

Well-estimated deep learning models, **BrainNetCNN** [20], **BrainGNN** [12] are also involved. Notably, BrainNetCNN was implemented and performed by concatenating the multi-modal brain networks by channel, as referenced in previous studies [21]. BrainGNN originally utilized partial Pearson and full correlation coefficients to build node features and adjacency matrices. We implemented BrainGNN for multi-modal studies by concatenating multi-modal brain networks as node features and using the partial correlation coefficient as the adjacency matrix.

Multimodal methods including **M-GCN** [11], **HGNN** (Hypergraph Graph Neural Network) [31], **DHGNN** (Dynamic Graph Hyper-Neural Network) [32], and the Triplet Attention Network (**TAN**) [35] are compared. These models were implemented with reference to the original architecture. The hyperparameters, such as the number of ROIs, were modified to suit our data. It is important to note that M-GCN and TAN are only available for two modalities, which are not applicable to the ABIDE datasets. Here, we use the Pearson correlation coefficient and partial correlation for evaluation.

Moreover, transductive GNNs are powerful tools for neuroimage studies with limited samples. **Population-GCN** and **MMP-GCN** [34] are implemented by building a population graph, where each node represents a participant. The adjacency matrix is constructed using both phenotype values including gender and age and the similarity between node features. Key features were selected by recursive feature elimination and vectorized into a set for each vertex and then concatenated. The number of selected features is determined through a search process with a step of 200.

## V. RESULTS

### A. Evaluation results on different tasks and datasets

1) *Classification on the progression of AD*: The quantitative results for the three classification sub-tasks of Task-1 (NC



TABLE III

CLASSIFICATION PERFORMANCE ON THE **TASK-1: PRECLINICAL DIAGNOSIS OF AD** IN TERMS OF ACCURACY (ACC %), SENSITIVITY (SEN %), AND SPECIFICITY (SPE %) AMONG ALL COMPETITIVE MODELS.

| Model               | Type        | NC vs MCI       |                   |                  | MCI vs AD       |                  |                   | NC vs AD        |                  |                  |
|---------------------|-------------|-----------------|-------------------|------------------|-----------------|------------------|-------------------|-----------------|------------------|------------------|
|                     |             | Acc ↑           | Sen ↑             | Spe ↑            | Acc ↑           | Sen ↑            | Spe ↑             | Acc ↑           | Sen ↑            | Spe ↑            |
| SVM                 | ML          | 52.2±9.3        | 52.4±9.0          | 52.2±10.6        | 60.0±9.9        | 59.8±14.9        | 60.8±7.8          | 75.4±9.8        | 77.5±14.0        | 75.3±8.8         |
| MLP                 | ML          | 70.1±8.8        | 75.0±15.8         | 73.1±13.6        | 69.1±5.4        | 73.6±9.3         | 70.1±9.9          | 77.8±6.5        | 76.6±6.8         | 80.4±8.3         |
| CP-MLP              | Tensor, ML  | 73.9±10.2       | 77.2±10.1         | 75.8±16.1        | 75.6±7.2        | 77.4±10.8        | 75.9±7.8          | 75.2±8.1        | 78.7±9.5         | 73.6±7.9         |
| TD-MLP              | Tensor, ML  | 73.9±7.6        | 73.9±9.5          | 76.6±9.5         | 73.2±5.9        | 79.8±12.7        | 70.7±5.4          | 75.9±10.9       | 82.6±14.7        | 73.2±10.6        |
| BrainNetCNN [20]    | CNN         | 72.1±7.1        | 75.9±9.3          | 70.4±10.8        | 71.8±5.0        | 74.4±6.8         | 70.8±5.0          | 81.3±11.3       | 81.0±11.7        | 82.1±10.1        |
| BrainGNN [12]       | GNN         | 76.0±5.9        | 81.1±9.9          | 71.5±12.6        | 74.3±4.8        | 76.6±5.2         | 73.5±12.8         | 82.0±2.4        | 84.6±5.6         | 76.8±6.1         |
| M-GCN [11]          | GNN         | 75.7±6.3        | 80.1±4.8          | 72.9±6.1         | 75.8±6.9        | 76.1±8.1         | 75.0±9.0          | 84.4±5.7        | 84.1±6.0         | 85.6±7.1         |
| HGNN [31]           | GNN         | 78.1±6.6        | 80.2±8.4          | 74.7±9.4         | 76.8±5.5        | 83.5±12.8        | 76.0±7.5          | 81.4±8.0        | 82.5±6.1         | 82.0±12.1        |
| DHGNN [32]          | GNN         | 79.2±6.0        | 83.0±9.9          | 77.6±7.3         | 75.8±4.2        | <u>85.4±12.9</u> | 73.9±8.4          | 82.5±4.1        | 86.4±10.7        | 82.1±11.4        |
| Population-GCN [54] | semi-GNN    | 75.5±5.9        | 82.5±10.0         | 71.6±8.0         | 75.5±5.9        | 80.2±14.4        | 74.0±5.4          | 82.0±7.5        | 85.9±7.5         | 75.9±8.0         |
| MMP-GCN [34]        | semi-GNN    | <u>80.6±7.6</u> | <u>86.2±10.2</u>  | <u>79.5±10.1</u> | 79.6±11.0       | 82.5±14.0        | <b>82.2±14.3</b>  | 84.4±11.1       | 88.1±12.4        | 85.1±13.2        |
| TAN [35]            | Attention   | 80.2±5.4        | 82.0±11.1         | 79.2±11.6        | 81.2±3.5        | 83.7±8.8         | <u>79.0±9.7</u>   | 85.9±4.5        | <u>89.3±12.9</u> | <u>86.3±10.3</u> |
| HC-GNN (Ours)       | Tensor, GNN | <b>81.5±4.1</b> | <b>90.6±10.0*</b> | <b>80.7±13.4</b> | <b>82.5±6.3</b> | <b>88.0±8.7*</b> | <b>82.2±12.6*</b> | <b>87.5±8.4</b> | <b>91.9±6.5*</b> | <b>88.0±11.3</b> |

\* indicates significant improvements (p-value < 0.05) compared with the second-best results.

TABLE IV

CLASSIFICATION PERFORMANCE ON THE **PSYCHIATRIC DISEASE DIAGNOSIS** IN TERMS OF ACCURACY (ACC %), SENSITIVITY (SEN %), AND SPECIFICITY (SPE %) AMONG ALL COMPETITIVE MODELS. † INDICATES THAT THE MODELS ARE TRAINED WITH TWO TYPES OF BRAIN NETWORKS ON THE ABIDE-I DATASET.

| Model               | Type        | NC vs ASD (ABIDE-I) |                 |                 | NC vs ADHD (HBN-ADHD) |                 |                 |
|---------------------|-------------|---------------------|-----------------|-----------------|-----------------------|-----------------|-----------------|
|                     |             | Acc ↑               | Sen ↑           | Spe ↑           | ACC ↑                 | Sen ↑           | Spe ↑           |
| SVM                 | ML          | 69.0±3.7            | 71.3±3.3        | 67.8±4.3        | 60.0±4.3              | 60.1±2.9        | 63.9±16.2       |
| MLP                 | ML          | 69.1±2.2            | 68.8±2.1        | 71.7±3.8        | 69.5±2.5              | 69.9±4.7        | 72.9±9.3        |
| CP-MLP              | Tensor, ML  | 71.1±3.1            | 69.4±4.7        | 72.3±4.4        | 72.9±5.4              | 73.9±5.7        | 74.2±11.8       |
| TD-MLP              | Tensor, ML  | 71.3±2.2            | 74.0±7.5        | 70.0±2.1        | 73.8±5.4              | 72.2±6.6        | 78.2±2.6        |
| BrainNetCNN [20]    | CNN         | 71.6±2.5            | 72.5±3.8        | <b>73.6±4.5</b> | 69.9±7.6              | 67.9±7.7        | <b>81.0±7.1</b> |
| BrainGNN [12]       | GNN         | 69.7±2.8            | 66.7±9.9        | 72.5±8.3        | 75.2±4.0              | <b>88.6±8.5</b> | 73.9±7.2        |
| M-GCN† [11]         | GNN         | 68.7±6.8            | 68.7±6.0        | 72.9±4.9        | 74.2±4.0              | 80.0±2.6        | 74.7±6.0        |
| HGNN [31]           | GNN         | 71.4±2.6            | 70.2±4.2        | 73.0±3.9        | 73.8±5.8              | 84.6±10.2       | 72.1±5.1        |
| DHGNN [32]          | GNN         | 71.9±3.3            | 72.1±3.5        | 72.0±4.4        | 72.5±4.0              | 85.6±8.0        | 70.2±4.2        |
| Population-GCN [54] | semi-GNN    | 69.9±1.4            | <b>79.0±5.4</b> | 68.0±1.8        | 76.2±10.1             | 79.3±7.1        | 73.1±9.8        |
| MMP-GCN [34]        | semi-GNN    | 70.5±3.9            | 71.1±9.6        | 72.4±7.9        | 76.7±5.1              | 76.1±6.3        | 75.6±3.7        |
| TAN† [35]           | Attention   | 71.7±2.8            | 71.6±2.7        | 69.5±5.3        | 78.6±5.0              | 83.7±8.1        | 75.7±5.8        |
| HC-GNN (Ours)       | Tensor, GNN | <b>73.6±3.6*</b>    | 76.2±4.4        | 73.1±7.2        | <b>81.6±6.2*</b>      | 87.3±4.8        | 78.7±8.2        |

\* indicates significant improvements (p-value < 0.05) compared with the second-best results.

vs. MCI, MCI vs. AD, and NC vs. AD) are presented in Table III, with the average and standard deviation across folds. Overall, tensor-based approaches (CP-MLP and TD-MLP) demonstrate promising performances across all three classification sub-tasks compared with MLP and SVM. These methods provide simple and efficient ways to model multi-modal data. In comparison, deep learning models, especially graph neural networks, consistently enhance classification performance, such as M-GCN and HGNN. This suggests that the structural topological properties of the brain are valuable for distinguishing the preclinical AD. Besides, the semi-supervised GNN method, MMP-GCN, outperforms most supervised GNN methods. Furthermore, when considering the three classification sub-tasks, our proposed HC-GNN emerges as the top-performing approach among all models, achieving accuracies of 81.5%, 82.5%, and 87.5% in distinguishing NC from MCI, MCI from AD, and NC from AD, respectively.

2) *Classification of the psychiatric disease diagnosis*: Table IV presents the comparison results for distinguishing ASD and ADHD from NC on the ABIDE-I and HBN datasets,

respectively. We observe that, similar to the ADNI dataset, GNN approaches generally perform the best in most cases. Interestingly, tensor-based methods achieve promising results and outperform some GNN models. For instance, on the ABIDE-I dataset, CP-MLP and TD-MLP exhibit better accuracy performances (CP-MLP: 71.1%, TD-MLP: 71.3%) compared to BrainGNN (69.7%), M-GCN (68.7%), and MMP-GCN (70.5%). On one hand, BrainGNN and M-GCN were available to handle two modalities, and their performance suffers when dealing with limited connectome information. On the other hand, although MMP-GCN benefits from semi-supervised learning on a transductive graph, it fails to capture inter-modal dependencies effectively. In contrast, our proposed HC-GNN and TAN are capable of modeling intersection representations between modalities, thereby exhibiting better performance. Furthermore, on the HBN-ADHD dataset, our proposed HC-GNN outperforms the second-best model, TAN, with a 3% improvement in accuracy. This is due to the fact that most compared models are designed for either fMRI connectomes or DTI connectomes and struggle to generalize



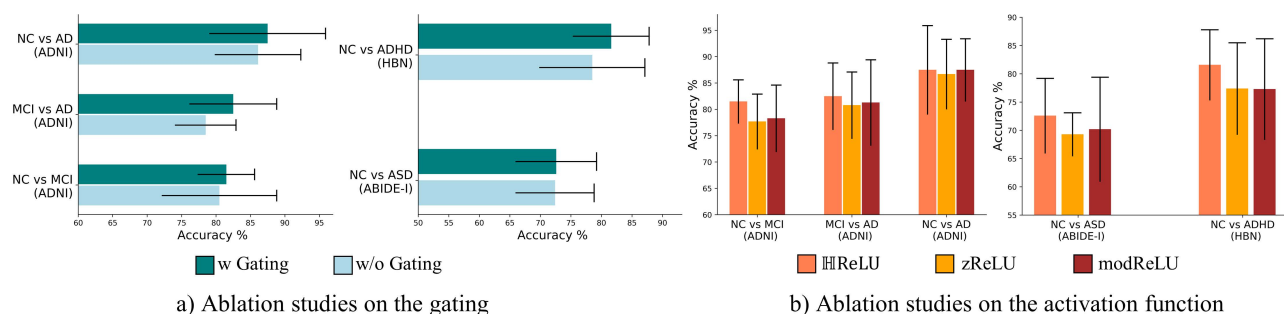


Fig. 3. Ablation studies on a) the gating mechanism and b) activation function on three datasets.

well to the learning of both fMRI and EEG. In contrast, our HC-GNN exhibits better scalability when learning from various types of brain networks. We suspect that this is caused by the deep intersection among modalities that facilitates generating high-order generalizations of matrices for multi-modal learning.

TABLE V  
ABLATION STUDIES ON THE HYPERCOMPLEX MODELING OF TENSOR-GRAPHS.

| Dataset      | ADNI            |                 | ABIDE-I         |                 | HBN             |
|--------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Task         | NC vs. MCI      | MCI vs. AD      | NC vs. AD       | NC vs. ASD      | NC vs. ADHD     |
| Real-valued  | 77.5±6.6        | 74.5±10.5       | 82.2±10.2       | 70.5±7.8        | 74.3±11.9       |
| Tensor       | 80.0±8.7        | 80.6±9.3        | 85.8±13.3       | 71.0±2.6        | 76.6±5.5        |
| Hypercomplex | <b>81.5±4.1</b> | <b>82.5±6.3</b> | <b>87.5±8.4</b> | <b>73.6±6.6</b> | <b>81.6±6.2</b> |

TABLE VI

MODEL GENERALIZABILITY OF CLASSIFICATION PERFORMANCE ON AN INDEPENDENT TESTING DATASET (ABIDE-II) IN TERMS OF ACCURACY (ACC %), SENSITIVITY (SEN %), AND SPECIFICITY (SPE %) AMONG ALL COMPETITIVE MODELS. † INDICATES THAT THE MODELS ARE TRAINED WITH TWO TYPES OF BRAIN NETWORKS ON THE ABIDE-II DATASET.

| Model               | Type        | NC vs ASD (ABIDE-II) |                 |                 |
|---------------------|-------------|----------------------|-----------------|-----------------|
|                     |             | Acc ↑                | Sen ↑           | Spe ↑           |
| MLP                 | ML          | 67.3±0.3             | 66.3±1.5        | 68.1±1.4        |
| CP-MLP              | Tensor, ML  | 64.1±0.5             | 59.1±0.9        | 67.2±0.9        |
| TD-MLP              | Tensor, ML  | 65.1±0.4             | 65.4±0.8        | 65.3±0.4        |
| BrainNetCNN [20]    | CNN         | 66.2±0.7             | 66.0±1.6        | 66.5±1.5        |
| BrainGNN [12]       | GNN         | 63.2±0.6             | 50.1±3.4        | 74.1±5.6        |
| M-GCN† [11]         | GNN         | 64.8±0.1             | 64.8±2.2        | 64.8±0.1        |
| HGNN [31]           | GNN         | 65.4±0.7             | 63.9±1.9        | 66.5±1.4        |
| DHGNN [32]          | GNN         | 64.6±0.7             | 61.8±0.8        | 66.3±1.1        |
| Population-GCN [54] | semi-GNN    | 66.0±0.3             | 66.2±4.0        | 66.3±0.4        |
| MMP-GCN [34]        | semi-GNN    | 66.3±0.2             | <b>68.6±0.4</b> | 59.6±0.2        |
| TAN† [35]           | Transformer | 67.1±0.2             | 66.3± 0.7       | 68.4±0.1        |
| HC-GNN (Ours)       | Tensor, GNN | <b>68.7±0.5</b>      | 66.9±2.7        | <b>68.6±1.7</b> |

## B. Ablation study

In this section, we present our ablation studies that aim to investigate the effectiveness of the gating mechanism, the modeling of HNN, and the selection of hypercomplex activation functions in the learning of brain connectomes. These studies provide valuable insights into the contribution of each component and their impact on the overall performance of our HC-GNN model.

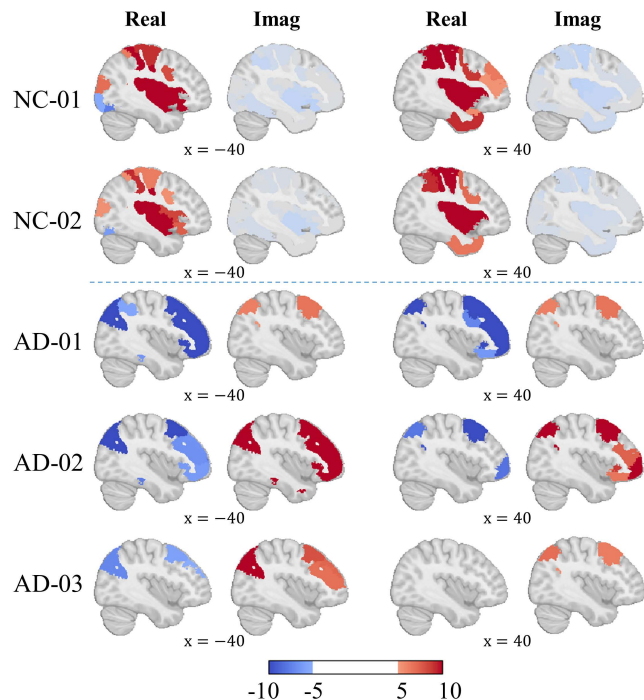


Fig. 4. Examples of the saliency maps for two NC and three AD subjects. The saliency values have been thresholded by 5 for better visualization.

1) *Gating Mechanism*: To examine the effectiveness of the gating mechanism, we conducted a comparison between our proposed HC-GNN models with and without the gating mechanism (referred to as "w Gating" and "w/o Gating" respectively). The results, depicted in Fig. 3, showcase the performance in terms of accuracy. It is evident that across all three tasks, our HC-GNN with the gating consistently outperforms the model without the gating, which indicates that the deep intersection of hypercomplex neural networks introduces unwanted inductive noise in coupling multi-modal brain networks.

2) *Activation Function*: Furthermore, we conducted an evaluation of the hypercomplex activation functions, including HReLU, zReLU, and modReLU. The comparative results presented in Fig. 3 demonstrate that the HReLU function consistently achieves the best performance across all tasks. This is reasonable for the hypercomplex neural networks, as it ensures compliance with the Cauchy-Riemann equations. We

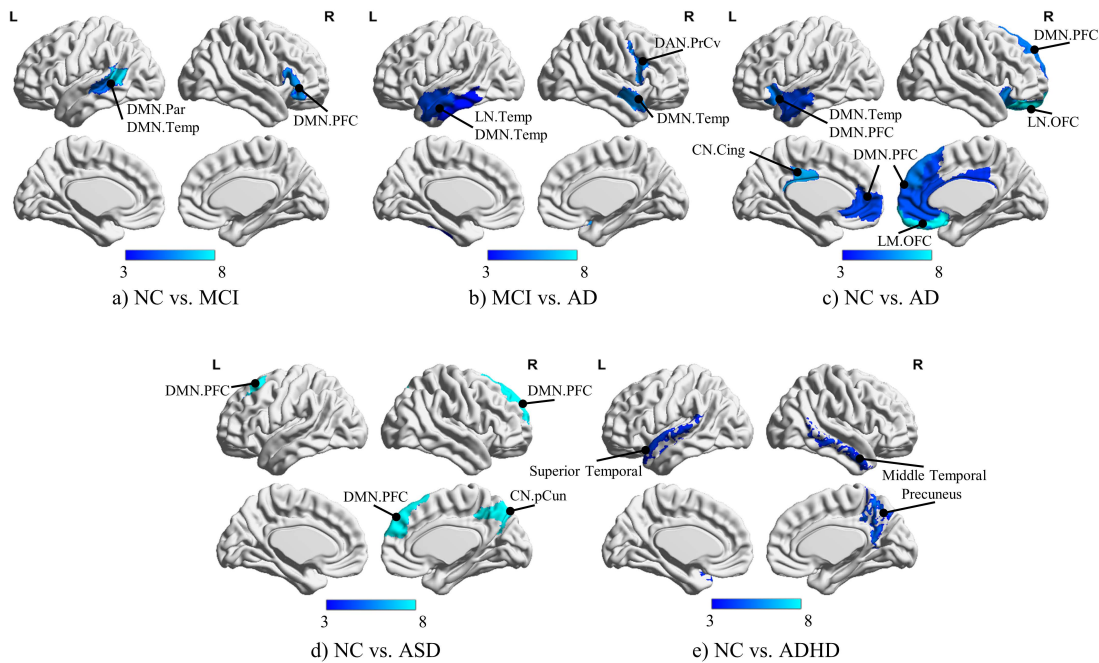


Fig. 5. Group-wise comparison on (a) NC vs. MCI, (b) MCI vs. AD, (c) NC vs. AD, (d) NC vs. ASD, and (e) NC vs. ADHD. Statistical results (F-statistic values are shown in blue with the cutoff ranging from 3 to 8) on the saliency maps with only significant regions (adjusted p-value < 0.05) are displayed. L: the left hemisphere. R: the right hemisphere. DMN: the default network. LN: the limbic network. CN: the control network. Temp: the temporal. Par: the parietal. Cing: the cingulate. PFC: the prefrontal cortex. PrCv: the precentral ventral. pCun: the precuneus.

hypothesize that modReLU and zReLU treat the hypercomplex inputs as a whole and filter the feature maps accordingly. In our study, however, the multi-modal inputs still retain their separate information within distinct imaginary units to some extent. Consequently, the  $\mathbb{H}$ ReLU function emerges as a more suitable choice for effectively mapping the heterogeneous modalities, leveraging the real and imaginary parts separately.

**3) Impact of Hypercomplex neural network:** In this study, our approach involves constructing tensor graphs from multi-modal data and employing hypercomplex neural networks to analyze these tensor graphs. A crucial aspect is to validate the effectiveness of using hypercomplex neural networks to model tensor graphs. To this end, we conducted ablation studies comparing two approaches: (1) Directly using multi-channel real-valued graph neural networks, and (2) extracting key features through Tucker Decomposition and feeding them into a multi-channel real-valued neural network. The multi-channel features are flattened before performing message passing on graphs. Table V presents the comparative results across all three tasks. It can be observed that the performances of the real-valued graph neural networks are on par with those of BrainNetCNN in Table III and Table IV, as both models capture inter-modal dependencies using multi-channels. However, Tucker Decomposition consistently improves performance, as it enables the extraction of key components and treats the multi-modal matrix as a unified entity. Nonetheless, our proposed HC-GNN consistently outperforms these methods.

### C. Generalization ability

To assess the model's generalization ability due to limited samples in neuroimaging and EEG datasets, we conducted test-

ing using the ABIDE-II dataset. The results of this evaluation are presented in Table VI, which includes the average and deviation results obtained from the 10-fold models trained on the ABIDE-I dataset. Comparing the results of the ABIDE-I dataset from Table IV, we can observe that the second-best model in the ABIDE-I dataset, BrainNetCNN, performs poorly in the ABIDE-II dataset. This indicates that BrainNetCNN has limited generalization ability for predicting ASD with different datasets. On the other hand, TAN, which utilizes self-attention to model multi-modal representations, demonstrates better generalization ability. This finding shows evidence that Transformer-like architecture is able to generalize to various fields. Compared with these models, our proposed HC-GNN achieves the best generalizability among all the models, outperforming the second best model with 1.6% improvements.

### D. Model Interpretability

To better understand which brain regions contribute most to the classification of the brain disorders, we analyzed the saliency maps to identify the brain regions that contribute most significantly.

Fig. 4 visualizes the saliency maps on the ADNI dataset as an example, where two NCs and three ADs participants are randomly sampled for comparison. For each participant, the saliency maps are averaged in the hidden dimension to align with the anatomical space. The saliency maps of the real and imaginary parts are displayed respectively, with red and blue indicating positive and negative values. The visualization reveals that the NCs exhibit similar pattern locations, while AD patients show distinctive key feature distributions. Moreover, there is a significant overlap in the key regions of

the real and imaginary parts, which can be attributed to the deep intersection of these features. Consequently, the readout function that maps hypercomplex numbers into real numbers plays a crucial role in extracting common features. Overall, compared with NC, the AD participants achieve higher values in the medial and dorsal prefrontal cortex, and the cingulate regions. These regions are found to be severely affected in AD found in previous works [55], [56]. This underscores the discrimination ability of our HC-GNN, demonstrating its capacity for diagnosis and to locate individual key patterns.

Moreover, we conducted group-wise statistical analysis on the saliency maps, as is shown in Fig. 5, where the F-statistic values of the regions found to be statistically significant (with a p-value of less than 0.05) are displayed. In the comparison between individuals with NC and those with MCI, three key regions were identified with significant differences. These regions include the left temporal lobe, the left parietal cortex, and the right prefrontal cortex (PFC) of the Default Mode Network (DMN). Besides the temporal lobe, in the distinction between AD and MCI, the right precentral ventral cortex of the Dorsal Attention Network (DAN) was also found to be discriminative. Additionally, in the comparison between NC and AD, significant differences were observed in the cingulate regions of both sides of the control Network (CN), the left temporal lobe of the DMN, and the left prefrontal cortex of the DMN. The temporal lobes play a crucial role in declarative memory and recognition. The posterior cingulate cortex (PCC) and medial temporal cortex are anatomically and functionally connected, and amyloid deposition in these regions contributes to synaptic dysfunction and eventual neuronal loss in the hippocampus and surrounding cortical structures, leading to memory impairment [57], [58]. Furthermore, AD is a neurodegenerative condition that affects heteromodal association regions, such as PFC. Dysfunction in PFC, which is involved in semantic memory processing, has been observed in AD patients, suggesting compromised semantic memory in this population [56], [59]. Our findings align with these studies and provide additional evidence supporting their conclusions.

Furthermore, significant differences were observed in the PFC of the Default Mode Network and the PCC of the Frontoparietal Network in comparisons between individuals with NC and ASD. Additionally, the right middle temporal cortex, the right precuneus, and the left superior temporal cortex are key regions in the comparison between NC and ADHD. These regions belong to the default mode network, which is involved in adaptive responses in both ASD and ADHD [60]–[62]. Our findings in the prefrontal and temporal cortex coincide with these studies. Furthermore, the right precuneus, which is part of the parietal cortex and involved in a wide range of cognitive processes, is found in both ASD and ADHD. Altered precuneus connectivity has been implicated in the neuropathology of ASD and ADHD [63]–[65]. Previous studies suggest that increased activation in these areas likely reflects problems in deactivating the default mode network [66]. Additionally, this finding may be related to the similar symptomatology observed in children with ASD and ADHD, indicating that the two conditions might exist on the same continuum and are likely to share many common

factors [4], [67], [68]. Our finding aligns with these studies and offers novel insights for examining and comparing the neural signatures of both conditions.

## VI. DISCUSSION

In this study, our proposed HC-GNN facilitates multimodal brain network studies with various types and numbers of modalities by using complex and hypercomplex representations. Compared with previous studies that couple two modalities in a fixed setting [5], [11], our method is able to generalize to different types of modalities within the same architecture, simultaneously coupling various modalities and relating their dependencies. Extensive experiments were conducted to show the superiority of HC-GNN in adaptability and generalizability to different conditions for brain network analysis (i.e., functional and structural coupling on the ADNI dataset, leveraging multiple types of brain networks to improve diagnosis on the ABIDE and HBN-ADHD datasets).

In addition, it's worth noting that the ADNI dataset comprises participants with both fMRI and DTI images, which are challenging to collect. Although the integration of functional and structural connectomes has greatly facilitated computer-aided diagnosis of neurodegenerative diseases, it comes with a high cost in terms of data acquisition in clinical scenarios. One approach to mitigate this challenge is to utilize Portable wearable devices such as EEG and fNIRS. The results from HBN-ADHD demonstrate that incorporating resting-state fMRI and EEG data is effective for diagnosing ADHD with high accuracy. EEG data is more easily accessible compared to MRI scans. Furthermore, our HC-GNN outperforms previous studies [28], [69] (around 71% accuracy) in terms of classification results for distinguishing between NC and ADHD. While these previous studies primarily rely on fMRI, which limits their performance, our HC-GNN achieves an accuracy of 81.6%. Similarly, the ABIDEI and ABIDE-II datasets demonstrated that the presence of three distinct functional brain network types significantly enhanced the generalizability and potential of the performance outcomes. Overall, our HC-GNN provides novel insights into modeling various types of brain networks with flexible extension capabilities and promising performance.

In addition to its strengths, our proposed HC-GNN also has some limitations. On the one hand, the architecture relies on spatial alignment to maintain consistency. Despite the application of feature engineering and selection techniques in various modalities within some research, these methods do not effectively address the interdependencies between modalities. This oversight restricts the clarity with which they can elucidate the connections between multimodal interactions and pathological conditions. To this end, spatial alignment is a prerequisite step for associating inter-modal relationships in multimodal brain network studies. On the other hand, the deep intersection operations involved in hypercomplex neural networks are time-consuming, even though they do reduce the number of parameters compared to standard convolutional layers. While this issue is acceptable for most brain network studies given their relatively small sample sizes, it remains



a time-consuming process for other multimodal cases such as images and language texts. In our future work, we plan to explore strategies for optimizing the training process and the architecture to improve efficiency in other multi-modal applications.

## VII. CONCLUSION

In this study, we propose a hypercomplex neural network, HC-GNN, for modeling multi-modal tensor graphs. HC-GNN leverages gating and dynamic graph mechanisms to attentively couple multi-modal representations in the hypercomplex domain. Our extensive experiments demonstrate the significant potential of HC-GNN in modeling multi-modal brain networks, showcasing its superior performance in brain network analysis, as well as its powerful generalizability and scalability across various modalities. Additionally, HC-GNN facilitates meaningful individual-level and group-level interpretable feature mapping, providing valuable insights for clinical guidance. Overall, HC-GNN holds great potential for multi-modal brain network studies and biomarker detection, highlighting its deep intersection of multi-modal brain networks for representation learning.

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